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Road lighting - Part 2: Performance requirements

Eclairage public - Partie 2: Exigences de performance

Straßenbeleuchtung - Teil 2: Güteermkmale

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Contents

page

Foreword	3
Introduction	4
1 Scope.....	5
2 Normative references.....	5
3 Terms and definitions.....	5
4 ME/MEW-series of lighting classes	7
5 CE-series of lighting classes	8
6 S-, A-, ES- and EV- series of lighting classes	9
7 Appearance and environmental aspects	11
Annex A (informative) Installed classes for glare restriction and control of obtrusive light.....	13
A.1 Luminous intensity classes	13
A.2 Glare index classes.....	13
Annex B (informative) Lighting of pedestrian crossings.....	15
Bibliography	16

Foreword

This document (EN 13201-2:2003) has been prepared by Technical Committee CEN/TC 169 "Light and lighting", the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by May 2004, and conflicting national standards shall be withdrawn at the latest by May 2004.

This document EN 13201-2 has been worked out by the Joint Working Group of CEN/TC 169 "Light and lighting" and CEN/TC 226 "Road Equipment", the secretariat of which is held by AFNOR.

Annexes A and B are informative.

This document includes a Bibliography.

This standard, EN 13201 *Road Lighting*, consists of three parts. This document is;

Part 2: Performance requirements

The other parts of EN 13201 are:

Part 3: Calculation of performance

Part 4: Methods of measuring lighting performance

According to the CEN/CENELEC Internal Regulations, the national standards organizations of the following countries are bound to implement this European Standard: Austria, Belgium, Czech Republic, Denmark, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Luxembourg, Malta, Netherlands, Norway, Portugal, Slovakia, Spain, Sweden, Switzerland and the United Kingdom.

Introduction

A lighting class is defined by a set of photometric requirements aiming at the visual needs of certain road users in certain types of road areas and environment.

The purpose of introducing lighting classes is to make it easier to develop and use road lighting products and services in CEN member countries. The lighting classes have been defined with consideration of road lighting standards in these countries aiming at harmonization of requirements where possible. However, some lighting classes and subclasses reflect particular situations and national approaches based on traditional, climatic or other conditions.

The ME classes are intended for drivers of motorized vehicles for use on traffic routes, and in some countries also residential roads, allowing medium to high driving speeds.

The CE classes are also intended for drivers of motorized vehicles, but for use on conflict areas such as shopping streets, road intersections of some complexity, roundabouts and queuing areas. These classes have applications also for pedestrians and pedal cyclists.

The S and A classes are intended for pedestrians and pedal cyclists for use on footways and cycleways, emergency lanes and other road areas lying separately or along the carriageway of a traffic route, residential roads, pedestrian streets, parking areas, schoolyards etc.

The ES classes are intended as an additional class in situations where public lighting is necessary for the identification of persons and objects and in road areas with a higher than normal crime risk.

The EV classes are intended as an additional class in situations where vertical surfaces need to be seen in such road areas as toll stations, interchange areas etc.

The requirements of the lighting classes reflect the category of road user in question or the type of road area. Thus the ME classes are based on the road surface luminance, while the CE, S and A classes are based on the illumination of the road area. The S and A classes reflect different priorities to the road lighting. The ES classes are based on semi-cylindrical illuminance, while the EV classes are based on the vertical plane illuminance.

The ME classes present increasingly stronger requirements in the order ME 6, ME 5,...ME 1 forming steps of the lighting level as measured for instance in illuminance. The other classes are arranged in the same way, and so that their steps interlock.

Environmental aspects of road lighting are considered in clause 7 in terms of day time appearance, night time appearance and light emitted in directions, where it is neither necessary nor desirable. The purpose is to point to matters that can be included in tender specifications or similar, when relevant.

Installed intensity classes for the restriction of disability glare and control of obtrusive light G.1, G.2, G.3, G.4, G.5 and G.6 are introduced in the informative annex A. The use of G classes is mentioned in clause 5 for conflict areas and in clause 7 on appearance and environmental aspects.

Installed glare index classes for the restriction of discomfort glare D.0, D.1, D.2, D.3, D.4, D.5 and D.6 are introduced in the informative annex A as well. These classes are intended mainly for road areas lighted for the benefit of pedestrians and pedal cyclists.

Local lighting of pedestrian crossings is considered in the informative annex B. The intention of local lighting is to attract the attention of drivers of motorized vehicles to the presence of the pedestrian crossing and to illuminate pedestrians in or at the crossing area.

1 Scope

This part of this European Standard defines, according to photometric requirements, lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.

NOTE Installed intensity classes for the restriction of disability glare and control of obtrusive light and installed glare index classes for the restriction of discomfort glare are defined in annex A.

2 Normative references

This European Standard incorporates by dated or undated reference, provisions from other publications. These normative references are cited at the appropriate places in the text, and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this European Standard only when incorporated in it by amendment or revision. For undated references the latest edition of the publication referred to applies (including amendments).

EN 13201-3, *Road lighting — Part 3: Calculation of performance*.

EN 13201-4, *Road lighting — Part 4: Methods of measuring lighting performance*.

3 Terms and definitions

For the purposes of this European Standard, the following terms and definitions apply.

3.1

average road surface luminance (of a carriageway of a road) ($\bar{L}$)

luminance of the road surface averaged over the carriageway

NOTE Unit is candelas per square metre (cd/m^2).

3.2

longitudinal uniformity (of road surface luminance of a driving lane)

ratio of the lowest to the highest road surface luminance found in a line in the centre along a driving lane

3.3

longitudinal uniformity (of road surface luminance of a carriageway) (U_l)

lowest of the longitudinal uniformities of the driving lanes of the carriageway

3.4

threshold increment (T_I)

measure of the loss of visibility caused by the disability glare of the luminaires of a road lighting installation

3.5

surround ratio (of illumination of a carriageway of a road) (SR)

average illuminance on strips just outside the edges of the carriageway in proportion to the average illuminance on strips just inside the edges

3.6

average illuminance (on a road area) ($\bar{E}_{hs}$)

horizontal illuminance averaged over a road area

NOTE Unit is lux (lx).

3.7

minimum illuminance (on a road area) (E_{min})

lowest illuminance on a road area

NOTE Unit is lux (lx).

3.8

hemispherical illuminance (at a point on a road area) (E_{hs})

luminous flux on a small hemisphere with a horizontal base. divided by the surface area of the hemisphere

NOTE Unit is lux (lx).

3.9

average hemispherical illuminance (on a road area) ($\bar{E}$)

hemispherical illuminance averaged over a road area

NOTE Unit is lux (lx).

3.10

overall uniformity (of road surface luminance, illuminance on a road area or hemispherical illuminance) (U_o)

ratio of the lowest to the average value

3.11

maintained level (of average road surface luminance, average or minimum illuminance on road area, average hemispherical illuminance, minimum semi-cylindrical illuminance or minimum vertical plane illuminance)

design level reduced by a maintenance factor to allow for depreciation

3.12

semi-cylindrical illuminance (at a point) (E_{sc})

total luminous flux falling on a curved surface of a very small semi-cylinder divided by the curved surface area of the semi-cylinder. The direction of the normal on the flat back area inside the semi-cylinder should be the direction of orientation of the semi-cylinder

NOTE Unit is lux (lx).

3.13

minimum semi-cylindrical illuminance (on a plane above a road area) ($E_{sc,min}$)

lowest semi-cylindrical illuminance on a plane at a height of 1,5 m above a road area

NOTE Unit is lux (lx).

3.14

vertical plane illuminance (at a point) (E_v)

illuminance on a vertical plane

NOTE Unit is lux (lx).

3.15**minimum vertical plane illuminance (on a plane above a road area) ($E_{v,min}$)**

lowest vertical plane illuminance on a plane at a specified height above the road area

NOTE Unit is lux (lx).

4 ME/MEW-series of lighting classes

The ME and MEW classes in Tables 1a and 1b are intended for drivers of motorized vehicles on traffic routes of medium to high driving speeds.

NOTE 1 Guidance on the application of these classes is given in prCEN/TR 13201-1.

The average road surface luminance ($\bar{L}$), the overall uniformity of the luminance (U_o), the longitudinal uniformity of the luminance (U_l), the threshold increment (Tl) and the surround ratio (SR) are to be calculated and measured in accordance with EN 13201-3 and EN 13201-4.

Table 1a — ME-series of lighting classes

Class	Luminance of the road surface of the carriageway for the dry road surface condition			Disability glare	Lighting of surroundings
	$\bar{L}$ in cd/m ² [minimum maintained]	U_o [minimum]	U_l [minimum]	Tl in % ^a [maximum]	SR ^{2b} [minimum]
ME1	2,0	0,4	0,7	10	0,5
ME2	1,5	0,4	0,7	10	0,5
ME3a	1,0	0,4	0,7	15	0,5
ME3b	1,0	0,4	0,6	15	0,5
ME3c	1,0	0,4	0,5	15	0,5
ME4a	0,75	0,4	0,6	15	0,5
ME4b	0,75	0,4	0,5	15	0,5
ME5	0,5	0,35	0,4	15	0,5
ME6	0,3	0,35	0,4	15	no requirement
^a An increase of 5 percentage points in Tl can be permitted where low luminance light sources are used. (see note 6)					
^b This criterion can be applied only where there are no traffic areas with their own requirements adjacent to the carriageway.					

NOTE 2 The road surface luminance is the result of the illumination of the road surface, the reflection properties of the road surface and the geometrical conditions of observation. Conventions are given in EN 13201-3 and EN 13201-4, aiming at driving along stretches of road with viewing distances of between 60 m and 180 m.

NOTE 3 The average luminance ($\bar{L}$) reflects the general luminance level at which the driver performs. At the low level of lighting used for road lighting, performance improves with luminance in terms of increasing contrast sensitivity, increasing visual acuity and amelioration of glare.

NOTE 4 The overall uniformity (U_o) measures in a general way the variation of luminances and indicates how well the road surface serves as a background for road markings, objects and other road users.

NOTE 5 The longitudinal uniformity (U_l) provides a measure of the conspicuity of the repeated pattern of bright and dark patches on the road. It relates to visual conditions on long uninterrupted sections of road.

NOTE 6 The threshold increment (T_l) indicates that although road lighting improves visual conditions it also causes disability glare to a degree depending on the type of luminaires, lamps and geometric situation. Low-pressure sodium lamps and fluorescent tubes are normally considered to be low luminance lamps. For these lamps, and luminaires providing less or equivalent luminance, footnote a of Table 1a and footnote b of Table 1b permits higher values.

NOTE 7 Lighting confined to the carriageway is inadequate for revealing the immediate surrounds of the road and revealing road users at the kerb. The requirements for the surround ratio (SR) apply only where there are no traffic areas with their own requirements adjacent to the carriageway, including footways, cycleways or emergency lanes.

Table 1b — MEW-series of lighting classes

Class	Luminance of the road surface of the carriageway for the dry and wet road surface condition			Disability glare	Lighting of surroundings	
	Dry condition			Wet		
	$\bar{L}$ in cd/m ² [minimum maintained]	U_0 [minimum]	U_l^a [minimum]	U_0 [minimum]	Tl in % ^b [maximum]	SR^c [minimum]
MEW1	2,0	0,4	0,6	0,15	10	0,5
MEW2	1,5	0,4	0,6	0,15	10	0,5
MEW3	1,0	0,4	0,6	0,15	15	0,5
MEW4	0,75	0,4	no requirement	0,15	15	0,5
MEW5	0,5	0,35	no requirement	0,15	15	0,5
^a The application of this criterion is voluntary, but it can be applied on motorways.						
^b An increase of 5 percentage points in Tl can be permitted where low luminance light sources are used. (see note 6)						
^c This criterion can only be applied where there are no traffic areas with their own requirements adjacent to the carriageway.						

NOTE 8 In some countries, the road surface is damp or wet for a significant part of the hours of darkness. For a selected wet condition, an additional requirement to the overall uniformity (U_0) can be made to apply to avoid a serious downgrading of the performance for some of the damp periods. The relevant table in that case is Table 1b.

5 CE-series of lighting classes

The CE classes in Table 2 are intended for drivers of motorized vehicles, and other road users, on conflict areas such as shopping streets, road intersections of some complexity, roundabouts, queuing areas etc.

NOTE 1 Guidance on the application of these classes is given in prCEN/TR 13201-1.

CE classes can also be applied to areas used by pedestrians and pedal cyclists, e.g. underpasses.

The average illuminance ($\bar{E}$) and the overall uniformity of the illuminance (U_0) are to be calculated and measured in accordance with EN 13201-3 and EN 13201-4.

The road area for which the requirements of Table 2 apply can include the carriageway only, when applying separate requirements for the adequate lighting of other road areas for pedestrian and cyclists, or it can include also other road areas.

NOTE 2 Limitation of glare can be achieved by the selection of luminaires according to the classes G.1, G.2, G.3, G.4, G.5 or G.6 of annex A. Alternatively, when it is practicable to evaluate T_I values for all relevant combinations of directions and observer positions, the T_I value taken from Table 1a can be applied.

Table 2 — CE-series of lighting classes

Class	Horizontal illuminance	
	$\overline{E}$ in lx [minimum maintained]	U_0 [minimum]
CE0	50	0,4
CE1	30	0,4
CE2	20	0,4
CE3	15	0,4
CE4	10	0,4
CE5	7,5	0,4

NOTE 3 The CE classes are mainly intended for use when the conventions for road surface luminance calculations do not apply or are impracticable. This can occur when the viewing distances are less than 60 m and when several observer positions are relevant. The CE classes are simultaneously intended for other road users on the conflict area. The CE classes have further application for pedestrian and pedal cyclists in such cases, where S and A classes defined in clause 6 are not adequate.

6 S-, A-, ES- and EV- series of lighting classes

The S classes in Table 3 or the A classes in Table 4 are intended for pedestrians and pedal cyclists on footways, cycleways, emergency lanes and other road areas lying separately or along the carriageway of a traffic route, and for residential roads, pedestrian streets, parking places, schoolyards etc.

The ES classes in Table 5 are intended as additional classes for pedestrian areas for the purposes of reducing crime and suppressing feelings of insecurity.

The EV classes in Table 6 are intended as additional classes in situations where vertical surfaces need to be seen, e.g. interchange areas.

NOTE 1 Guidance on the application of the above-mentioned classes is given in prCEN/TR 13201-1.

The average illuminance ($\overline{E}$), the minimum illuminance ($E_{\min}$), the average hemispherical illuminance ($\overline{E}_{\text{hs}}$), the overall uniformity of the hemispherical illuminance (U_0), the minimum semi-cylindrical illuminance ($E_{\text{sc,min}}$) and the minimum vertical plane illuminance ($E_{\text{v,min}}$) are to be calculated and measured in accordance with EN 13201-3 and EN 13201-4.

The road area for which the requirements of Tables 3, 4, 5 and 6 apply can include all the road area such as carriageways on residential roads and reserves between carriageways, footways and cycleways.

NOTE 2 Limitation of glare can be achieved by the selection of luminaires according to the classes D.1, D.2, D.3, D.4, D.5 or D.6 of annex A. For the ES classes of Table 5, only the classes D.1, D.2 or D.3 are relevant.

Table 3 — S-series of lighting classes

Class	Horizontal illuminance	
	$\overline{E}$ in lx ^a [minimum maintained]	$E_{\min}$ in lx [maintained]
S1	15	5
S2	10	3
S3	7,5	1,5
S4	5	1
S5	3	0,6
S6	2	0,6
S7	performance not determined	performance not determined
^a To provide for uniformity, the actual value of the maintained average illuminance may not exceed 1,5 times the minimum $\overline{E}$ value indicated for the class.		

Table 4 — A-series of lighting classes

Class	Hemispherical illuminance	
	$\overline{E}_{hs}$ in lx [minimum maintained]	U_0 [minimum]
A1	5	0,15
A2	3	0,15
A3	2	0,15
A4	1,5	0,15
A5	1	0,15
A6	performance not determined	performance not determined

Table 5 — ES-series of lighting classes

Semi-cylindrical illuminance	
Class	$E_{sc,min}$ in lx [maintained]
ES1	10
ES2	7,5
ES3	5
ES4	3
ES5	2
ES6	1,5
ES7	1
ES8	0,75
ES9	0,5

Table 6 — EV-series of lighting classes

Vertical plane illuminance	
Class	$E_{v,min}$ in lx [maintained]
EV1	50
EV2	30
EV3	10
EV4	7,5
EV5	5
EV6	0,5

7 Appearance and environmental aspects

The design and siting of road lighting installations and equipment can make a great difference to the appearance of the road and the road environment, by day and by night. This applies not only to the road user, but also to the observer viewing the installation from some distance off the road.

Consideration shall be given to the following matters related to:

Day time appearance:

- choice of supporting method, for example columns with or without brackets, suspension wires, or direct mounting on buildings;
- design and colour of lighting columns;
- scale and height of lighting columns or other suspension elements in relation to the height of adjacent buildings, trees and other salient objects in the field of view;
- location of lighting columns in relation to views of scenic value;
- design, length and tilt of brackets on columns;

- tilt of luminaire;
- choice of luminaire;

Night time appearance and comfort:

- colour appearance of the light;
- colour rendering of the light;
- mounting height of the luminaire;
- lit appearance of the luminaire;
- lit appearance of the complete installation;
- optical guidance by direct light from the luminaire;
- reduction of light levels in periods.

Minimizing light emitted in directions where it is neither necessary nor desirable:

- in rural or suburban conditions, the obtrusive view of road lighting installations seen at a distance across open country;
- light intruding into properties;
- light when emitted above the horizontal which when scattered in the atmosphere obscures the natural sight of the stars and impairs astronomical observation. Light emitted above the horizontal can be regulated by restriction of the upward light output ratio.

Annex A (informative)

Installed classes for glare restriction and control of obtrusive light

A.1 Luminous intensity classes

In some situations it can be necessary to restrict disability glare from installations where the threshold increment (T) cannot be calculated.

It can also be decided to control obtrusive light after consideration of the aspects listed in clause 7.

Table A.1 gives installed luminous intensity classes G.1, G.2, G.3, G.4, G.5 and G.6 from which a class can be chosen to meet appropriate requirements for restriction of disability glare and/or the control of obtrusive light.

Table A.1 — Luminous intensity classes

Class	Maximum luminous intensity in cd / klm			Other requirements
	at 70° ^a	at 80° ^a	at 90° ^a	
G1		200	50	None
G2		150	30	None
G3		100	20	None
G4	500	100	10	Luminous intensities above 95° ^a to be zero
G5	350	100	10	Luminous intensities above 95° ^a to be zero
G6	350	100	0	Luminous intensities above 90° ^a to be zero

^a Any direction forming the specified angle from the downward vertical, with the luminaire installed for use.

NOTE 1 The threshold increment (T) measures the veiling luminance caused by disability glare in relation to the average road surface luminance, approximately in proportion. The lamp output affects both terms equally and, therefore, the intensities in proportion to the lamp output are used in Table A.1.

NOTE 2 G.1, G.2 and G.3 correspond to "semi cut-off" and "cut-off" concepts of traditional use, with requirements, however, modified to suit the prevailing use of light sources and luminaires. G.4, G.5 and G.6 correspond to full cut-off.

A.2 Glare index classes

Table A.2 gives glare index classes D.0, D.1, D.2, D.3, D.4, D.5 and D.6 from which a class can be chosen to meet appropriate requirements for restriction of discomfort glare.

The glare index is $I \times A^{-0.5}$, unit cd/m, where:

- I is the maximum value of the luminous intensity (cd) in any direction forming an angle of 85° from the downward vertical

- A is the apparent area (m^2) of the luminous parts of the luminaire on a plane perpendicular to the direction of I . If in the direction of I , parts of the light source are visible, either directly or as images, class D.0 applies.

NOTE 1 The apparent area A , as determined in a horizontal direction, can be sufficiently accurate.

NOTE 2 When a light distribution table for the luminaire in the conditions of operation is available, I can be derived from the table.

Table A.2 — Glare index classes

Class	D0	D1	D2	D3	D4	D5	D6
Glare index maximum	—	7000	5500	4000	2000	1000	500

EXAMPLE 1 A luminous sphere has a diameter of 0,6 m and an intensity of 60 cd per 1000 lm bare lamp output in any direction. The apparent area is $\pi \times 0,6^2/4 \text{ m}^2 = 0,28 \text{ m}^2$ and the glare rating value is $60 \times 0,28^{-0,5} = 113$ per 1000 lm lamp output. With current lamp output values, the use of 50 W or 70 W high-pressure sodium lamps for this particular luminaire results in classes D.6 or D.5 respectively, while the use of 50 W, 80 W or 125 W mercury lamps results in classes D.6, D.6 or D.5 respectively.

EXAMPLE 2 A luminaire has a suspended bowl, shaped like a box of height 0,1 m, length 0,3 m and width 0,2 m. The maximum value of $I \times A^{-0,5}$ at 85° to the downward vertical is found in the plane perpendicular to the longitudinal direction of the bowl, where

I is 50 cd per 1000 lm lamp output;

A is $0,3 \times 0,2 \times \cos 85^\circ + 0,3 \times 0,1 \times \cos (90^\circ - 85^\circ) \text{ m}^2 = 0,035 \text{ m}^2$, so that the glare rating value is $50 \times 0,035^{-0,5} = 267$ per 1000 lm lamp output.

Annex B

(informative)

Lighting of pedestrian crossings

Pedestrian crossings can require special consideration. In some countries national standards exist which give further guidance relative to national practices.

When a sufficiently high road surface luminance level can be provided, it can be possible to position the normal road lighting luminaires so as to provide good negative contrast with the pedestrian visible as a dark silhouette against a bright background.

Local lighting with additional luminaires is considered in the following. The intention is to directly illuminate pedestrians on or at the crossing and to draw the attention of drivers of motorized vehicles to the presence of the crossing.

The type of the additional luminaires and their positions and orientations relative to the crossing area should be considered so as to achieve positive contrast, and so as not to cause undue glare to drivers. One solution is to mount luminaires a short distance before the crossing in the direction of approaching motorized traffic, and direct the light onto the side of pedestrians facing the drivers of this traffic. For a two-way road, a luminaire is mounted before the crossing in each direction of the traffic flow on the side of the road on which traffic drives. Luminaires with asymmetric light output are suitable, causing less glare to drivers.

Local lighting can be arranged to provide a sufficient illumination of the pedestrian on the side turning towards the traffic at all locations of the road crossing area. The illuminance, when measured on a vertical plane, should be significantly higher than the horizontal illuminance produced by road lighting on the carriageway of the road. Zones at either end of the road crossing, where pedestrians wait to enter the crossing, should receive adequate illumination. Lighting confined to a narrow band around the crossing area produces a dramatic effect assisting in raising attention.

Bibliography

prCEN/TR 13201-1, *Road lighting — Selection of lighting classes*.

EN 12665, *Light and lighting — Basic terms and criteria for specifying lighting requirements*.

IEC 60050-845, *International Electrotechnical Vocabulary — Chapter 845: Lighting*.

NOTE CIE Publication 17.4 International Lighting Vocabulary is identical to IEC 60050-845.

CIE Publication 115:1995, *Recommendations for the lighting of roads for motor and pedestrian traffic*.